



High-Dimensional Statistics

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Assignment 3

Instructions

- **Due.** December 27, 2025
- **Scoring.** The assignment contains **140 points** in total, but **100 points** are sufficient to receive full credit. Extra points will not be carried over to other assignments.
- **Late policy.** A maximum of three assignments may be submitted late, each with a delay of up to 3 days. Submissions beyond this limit or after the allowed delay will not be accepted.
- **Collaboration.** If you discuss concepts with classmates, list the collaborators.
- State assumptions, cite any less well-known lemmas or results you use, and keep proofs concise.

Problem 1. Mixture distributions and KL divergence

[10 points]

Given a collection of distributions $\{\mathbb{P}_1, \dots, \mathbb{P}_M\}$, consider the mixture distribution $\bar{\mathbb{Q}} = \frac{1}{M} \sum_{j=1}^M \mathbb{P}_j$. Show that

$$\frac{1}{M} \sum_{j=1}^M D(\mathbb{P}_j \| \bar{\mathbb{Q}}) \leq \frac{1}{M} \sum_{j=1}^M D(\mathbb{P}_j \| \mathbb{Q})$$

for any other distribution $\mathbb{Q}$.

Problem 2. f -divergences

[15 points]

Let $f : \mathbb{R}_+ \rightarrow \mathbb{R}$ be a strictly convex function. Given two distributions $\mathbb{P}$ and $\mathbb{Q}$ (with densities p and q , respectively), their f -divergence is given by

$$D_f(\mathbb{P} \| \mathbb{Q}) := \int q(x) f(p(x)/q(x)) \nu(dx). \quad (1)$$

- Show that the Kullback–Leibler divergence corresponds to the f -divergence defined by $f(t) = t \log t$.
- Compute the f -divergence generated by $f(t) = -\log(t)$.
- Show that the squared Hellinger divergence $H^2(\mathbb{P} \| \mathbb{Q})$ is also an f -divergence for an appropriate choice of f .
- Compute the f -divergence generated by the function $f(t) = 1 - \sqrt{t}$.

Problem 3.

[10 points]

Let $\mathbb{P}_0$ and $\mathbb{P}_1$ be two probability measures. Prove that for any test ψ , it holds

$$\max_{j=0,1} \mathbb{P}_j(\psi \neq j) \geq \frac{1}{4} e^{-\text{KL}(\mathbb{P}_0, \mathbb{P}_1)}.$$

Problem 4. Binary Testing and Le Cam Two-Point Method

[20 points]

Let $X_1, \dots, X_n$ be i.i.d. $\mathcal{N}(\theta, 1)$ with unknown mean $\theta \in [-1, 1]$. We estimate θ under squared-error loss, and define the minimax risk

$$R_n^* := \inf_{\hat{\theta}} \sup_{\theta \in [-1, 1]} \mathbb{E}_{\theta}[(\hat{\theta} - \theta)^2].$$

1. Consider the binary testing problem

$$H_0 : \theta = -\delta \quad \text{vs.} \quad H_1 : \theta = +\delta,$$

where $\delta > 0$ is such that $\pm\delta \in [-1, 1]$. Let P_0 and P_1 denote the joint distributions of $(X_1, \dots, X_n)$ under $\theta = -\delta$ and $\theta = +\delta$, respectively. Show that the Kullback–Leibler divergence satisfies

$$D(P_0 \parallel P_1) = D(P_1 \parallel P_0) = 2n\delta^2.$$

2. Use Pinsker's inequality

$$\|P_0 - P_1\|_{\text{TV}}^2 \leq \frac{1}{2} D(P_0 \parallel P_1)$$

to prove that there exists a constant $C > 0$ such that

$$\|P_0 - P_1\|_{\text{TV}} \leq C\sqrt{n}\delta.$$

3. Le Cam's two-point method states that for any estimator $\hat{\theta}$,

$$\sup_{\theta \in \{-\delta, +\delta\}} \mathbb{E}_\theta [(\hat{\theta} - \theta)^2] \geq \frac{(\theta_1 - \theta_0)^2}{4} (1 - \|P_0 - P_1\|_{\text{TV}}),$$

where $\theta_0 = -\delta$ and $\theta_1 = +\delta$. Choose $\delta = \delta_n$ as a function of n so that $\|P_0 - P_1\|_{\text{TV}} \leq 1/2$ for all large n , and deduce that

$$R_n^* = \inf_{\hat{\theta}} \sup_{\theta \in [-1, 1]} \mathbb{E}_\theta [(\hat{\theta} - \theta)^2] \geq \frac{c}{n} \quad \text{for all sufficiently large } n,$$

for some explicit constant $c > 0$ that you should determine.

Problem 5. Divergences and Distances between Probability Measures [25 points]

Let $\mathcal{X}$ be a finite set, and let P and Q be two probability measures on $\mathcal{X}$ with $P(x) > 0$ and $Q(x) > 0$ for all $x \in \mathcal{X}$. Recall the definitions of the following divergences and distances between P and Q :

$$\text{Total variation distance:} \quad \|P - Q\|_{\text{TV}} := \frac{1}{2} \sum_{x \in \mathcal{X}} |P(x) - Q(x)|,$$

$$\text{Kullback–Leibler divergence:} \quad D(P \parallel Q) := \sum_{x \in \mathcal{X}} P(x) \log \frac{P(x)}{Q(x)},$$

$$\text{Squared Hellinger distance:} \quad H^2(P, Q) := \frac{1}{2} \sum_{x \in \mathcal{X}} (\sqrt{P(x)} - \sqrt{Q(x)})^2,$$

$$\chi^2\text{-divergence:} \quad \chi^2(P \parallel Q) := \sum_{x \in \mathcal{X}} \frac{(P(x) - Q(x))^2}{Q(x)}$$

Problem. Prove or disprove each of the following statements, or show under which additional conditions they hold. You must give rigorous arguments and clearly indicate where you use inequalities such as Cauchy–Schwarz, Hölder, Pinsker, or Taylor expansions.

- (a) **TV vs. Hellinger.**

(a) Show that

$$H^2(P, Q) = 1 - \sum_{x \in \mathcal{X}} \sqrt{P(x)Q(x)}.$$

(b) Prove that

$$H^2(P, Q) \leq \|P - Q\|_{\text{TV}} \leq \sqrt{2} H(P, Q).$$

In particular, deduce that $H(P, Q) \rightarrow 0$ if and only if $\|P - Q\|_{\text{TV}} \rightarrow 0$.

(b) TV vs. χ^2 and KL.

(a) Use Cauchy–Schwarz to show that

$$\|P - Q\|_{\text{TV}}^2 \leq \frac{1}{4} \chi^2(P \parallel Q).$$

(b) Show that

$$D(P \parallel Q) \geq \log\left(1 + \frac{1}{2} \chi^2(P \parallel Q)\right),$$

and deduce a bound of the form

$$\|P - Q\|_{\text{TV}}^2 \leq C_1 D(P \parallel Q),$$

for some universal constant $C_1 > 0$ that you should make explicit. Compare your constant with the constant in Pinsker’s inequality

$$\|P - Q\|_{\text{TV}}^2 \leq \frac{1}{2} D(P \parallel Q).$$

(c) KL vs. Hellinger.

(a) Show that for all $u > 0$,

$$\log u \leq (u - 1) - \frac{(u - 1)^2}{2u}.$$

(Hint: study the function $f(u) = (u - 1) - \frac{(u-1)^2}{2u} - \log u$ using calculus.)

(b) Apply the inequality in (c.1) to $u = \frac{P(x)}{Q(x)}$ and sum over x to prove that

$$D(P \parallel Q) \leq 2 H^2(P, Q) \left(1 + \chi^2(P \parallel Q)\right).$$

Discuss in which regimes (e.g. P close to Q) this upper bound is nontrivial and what it implies about the local equivalence of D and H^2 .

Problem 6. Le Cam’s inequality

[10 points]

Prove the upper bound

$$\|\mathbb{P} - \mathbb{Q}\|_{\text{TV}} \leq H(\mathbb{P} \parallel \mathbb{Q}) \sqrt{1 - \frac{H^2(\mathbb{P} \parallel \mathbb{Q})}{4}}. \quad (15.10)$$

on the total variation norm in terms of the Hellinger distance. (*Hint*: The Cauchy–Schwarz inequality could be useful.)

Problem 7. Bounds for Gaussian location family

[20 points]

Consider the Gaussian location family. For a fixed variance σ^2 , let $\mathbb{P}_\theta$ be the distribution of a normal random variable $\mathcal{N}(\theta, \sigma^2)$. We observe a collection $Z = (Y_1, \dots, Y_n)$ of n i.i.d. samples drawn from $\mathbb{P}_\theta$.

Our goal is to estimate the mean θ under two different loss functions:

- Absolute error: $L(\hat{\theta}, \theta) = |\hat{\theta} - \theta|$
- Squared error: $L(\hat{\theta}, \theta) = (\hat{\theta} - \theta)^2$

(a) Standard Le Cam Bound

Use the two-point form of Le Cam's method to derive the following minimax lower bounds for the absolute and squared error:

$$\inf_{\hat{\theta}} \sup_{\theta \in \mathbb{R}} \mathbb{E}_\theta[|\hat{\theta} - \theta|] \geq \frac{1}{12} \frac{\sigma}{\sqrt{n}} \quad \text{and} \quad \inf_{\hat{\theta}} \sup_{\theta \in \mathbb{R}} \mathbb{E}_\theta[(\hat{\theta} - \theta)^2] \geq \frac{1}{24} \frac{\sigma^2}{n}$$

(b) Sharper Bounds via Pinsker's Inequality

The constants obtained in Part (a) are not optimal. Using the Pinsker-Csiszár-Kullback inequality, re-derive the lower bounds to obtain the sharper constants:

$$\inf_{\hat{\theta}} \sup_{\theta \in \mathbb{R}} \mathbb{E}_\theta[|\hat{\theta} - \theta|] \geq \frac{1}{8} \frac{\sigma}{\sqrt{n}} \quad \text{and} \quad \inf_{\hat{\theta}} \sup_{\theta \in \mathbb{R}} \mathbb{E}_\theta[(\hat{\theta} - \theta)^2] \geq \frac{1}{16} \frac{\sigma^2}{n}$$

Problem 8. Normal location model via Fano method

[15 points]

Recall from Problem 7 the normal location family, and the problem of estimating $\theta \in \mathbb{R}$ under the squared error. There we showed how to lower bound the minimax error using Le Cam's method; here let us derive a similar lower bound using Fano's method. Use Fano's method to get

$$\sup_{\theta \in \mathbb{R}} \mathbb{E}_\theta[(\hat{\theta} - \theta)^2] \geq \frac{1}{80} \frac{\sigma^2}{n}.$$

Problem 9. Minimax risk for sparse linear regression

[15 points]

Consider the high-dimensional linear regression model $y = \mathbf{X}\theta^* + w$, where the noise w is Gaussian and the vector θ^* is known *a priori* to be s -sparse (at most s non-zero entries). We wish to find the minimax lower bound over the set of sparse vectors in the unit ball:

$$\mathbb{S}^d(s) := \mathbb{B}_0^d(s) \cap \mathbb{B}_2(1) = \{\theta \in \mathbb{R}^d \mid \|\theta\|_0 \leq s, \|\theta\|_2 \leq 1\}$$

Use Fano's Method to derive the minimax lower bound on the squared error risk. specifically:

1. Construct a packing set of $\mathbb{S}^d(s)$ with log-cardinality $\log M \gtrsim s \log \frac{d}{s}$.
2. Use the mutual information bound (or KL divergence bound) to show that the minimax risk is lower bounded by:

$$\mathfrak{M}(\mathbb{S}^d(s); \|\cdot\|_2) \gtrsim \frac{\sigma^2}{n} s \log \frac{d}{s}.$$